Faculty of Engineering and Technology

Electrical and Computer Engineering Department

ENCS5343

Computer Vision - Final Project Report

Handwritten Arabic Numerals Recognition (0–9)

Prepared by:

Sadeen Ryahi – 1211109
Instructor: Dr.Aziz Qaroush

Sec: 1

Date: March 8, 2026

Abstract

Handwritten digit recognition is an important task in computer vision and pattern recognition with applications in document analysis and automated data entry. In this project, a Convolutional Neural Network (CNN) is developed to recognize handwritten Arabic digits (0–9). The dataset consists of grayscale images organized into ten classes representing the digits.

Before training, the images are preprocessed by normalizing pixel values and reshaping them to match the input format required by convolutional neural networks. The proposed CNN architecture includes multiple convolutional and pooling layers followed by fully connected layers for classification.

Experimental results show that the trained model achieves an accuracy of approximately 94% on the test dataset. The confusion matrix and performance metrics demonstrate that the model effectively learns discriminative features for recognizing handwritten digits. These results highlight the effectiveness of CNN-based approaches for image classification tasks.

1 Introduction

The automatic recognition of handwritten digits is a fundamental problem in computer vision and pattern recognition. Unlike printed text, handwritten digits vary significantly in stroke thickness, orientation, size, and individual writing styles, making accurate recognition challenging.

Recent advances in machine learning, particularly deep learning, have significantly improved the performance of handwritten digit recognition systems by allowing models to automatically learn relevant features directly from image data.

Convolutional Neural Networks (CNNs) are widely used for image classification tasks because they can effectively capture spatial patterns and hierarchical features in images.

In this project, a dataset of handwritten Arabic digits (0–9) is used to train and evaluate a CNN-based recognition system. The dataset contains grayscale images representing different handwritten styles of digits.

The objective of this project is to design and evaluate a CNN-based model capable of accurately classifying handwritten Arabic digits. The implemented system includes data preprocessing, CNN architecture design, model training, and performance evaluation using accuracy metrics and visualization techniques.

2 Methodology

2.1 Dataset

The dataset used in this project consists of handwritten Arabic digit images representing the numbers from 0 to 9. The images are organized into ten folders, where each folder corresponds to one digit class. All images are converted to grayscale and resized to a resolution of 28×28 pixels.

In total, the dataset contains 9,350 images with different handwriting styles. The dataset was divided into training and testing sets using an 80/20 split, resulting in 7,480 training images and 1,870 testing images.

2.2 Data Preprocessing

Before training the model, two preprocessing steps were applied to the input images.

First, pixel values were normalized from the original range $[0, 255]$ to $[0, 1]$ by dividing each pixel value by 255.0. This normalization improves numerical stability and accelerates the learning process.

Second, the image tensors were reshaped from $(N, 28, 28)$ to $(N, 28, 28, 1)$ to match the input format required by convolutional neural networks.

2.3 CNN Architecture

A Convolutional Neural Network (CNN) was designed using a sequential architecture consisting of convolutional, pooling, and fully connected layers.

The first convolutional layer applies 32 filters of size 3×3 with ReLU activation, followed by a 2×2 MaxPooling layer. The second convolutional layer uses 64 filters with the same kernel size, followed by another MaxPooling layer. A third convolutional layer with 128 filters is then applied to extract deeper image features.

The extracted feature maps are flattened and passed to a fully connected dense layer with 128 neurons and ReLU activation. A Dropout layer with a rate of 0.5 is added to reduce overfitting before the final output layer.

The output layer contains 10 neurons with a Softmax activation function that produces probability scores for each digit class.

The model was trained using the Adam optimizer with sparse categorical crossentropy as the loss function. Training was performed for 20 epochs with a batch size of 32, while 10% of the training data was used for validation during training.

The overall structure of the proposed CNN model is summarized in Table 1.

Layer	Type	Parameters
1	Conv2D	32 filters, 3×3, ReLU
2	MaxPooling	2×2
3	Conv2D	64 filters, 3×3, ReLU
4	MaxPooling	2×2
5	Conv2D	128 filters, 3×3, ReLU
6	Flatten	—
7	Dense	128 neurons, ReLU
8	Dropout	rate = 0.5
9	Output	10 neurons, Softmax

Table 1: CNN Architecture Summary

3 Experimental Results

3.1 Training Results

The CNN model was trained for 20 epochs using the training dataset, while 10% of the training data was used as validation during training. During the training process, the model gradually improved its performance as the training accuracy increased and the loss decreased over the epochs.

The dataset used in this project contains 9,350 handwritten digit images. After preprocessing, the dataset was split into 7,480 training images and 1,870 testing images.

The training curves show that the model quickly learned meaningful features from the images. Both training and validation accuracy increased steadily during training, indicating effective learning.

The final evaluation on the test dataset achieved an accuracy of **94.22%**. These results demonstrate that the CNN model is able to generalize well to unseen handwritten digits.

3.2 Accuracy and Loss Curves

To better understand the training behavior, accuracy and loss curves were plotted for both the training and validation datasets.

The accuracy curve increases rapidly during the early epochs and gradually stabilizes as training progresses. At the same time, the loss curve decreases consistently, indicating that the model is learning more accurate representations of the handwritten digits.

The relatively small gap between training and validation curves suggests that the model does not suffer from significant overfitting and maintains good generalization performance.

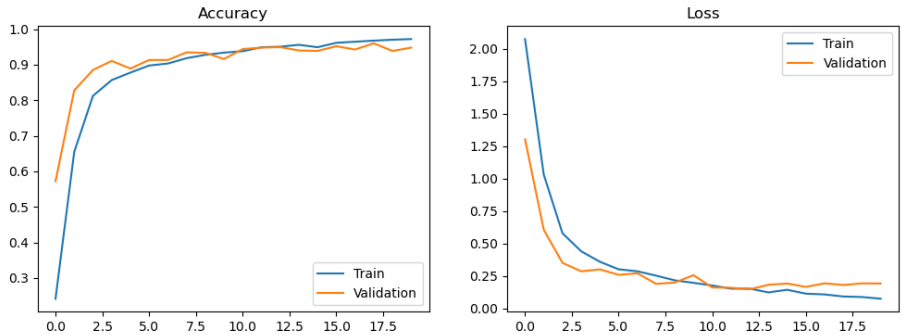

Figure 1: Training and validation accuracy and loss curves during model training

3.3 Confusion Matrix Analysis

A confusion matrix was computed using predictions on the test dataset. The matrix compares the predicted labels with the true digit classes, where diagonal elements represent correct classifications.

Most predictions lie along the diagonal of the matrix, indicating that the model correctly classifies the majority of the handwritten digits. Some minor misclassifications occur between visually similar digits, which is common in handwritten digit recognition tasks.

Overall, the confusion matrix confirms that the trained CNN model successfully learned discriminative features for distinguishing different digit classes.

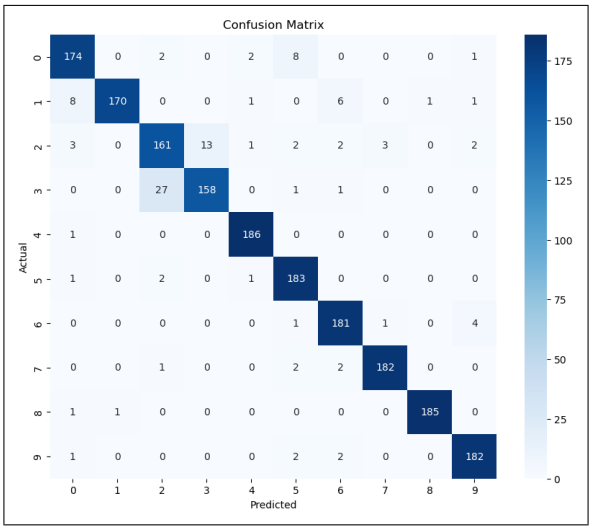

Figure 2: Confusion Matrix for the Test Dataset

4 Discussion

The experimental results demonstrate that the proposed CNN architecture is capable of effectively recognizing handwritten Arabic digits. The model achieved an overall test accuracy of approximately **94.22%**, indicating that convolutional neural networks are well suited for digit recognition tasks.

The training curves show a steady increase in accuracy and a consistent decrease in loss during the training process. This behavior indicates that the model successfully learned meaningful visual features from the input images.

The confusion matrix further confirms the effectiveness of the model, as most predictions are located along the diagonal, representing correct classifications. A small number of errors occur between visually similar digits, which is common in handwritten digit recognition tasks.

Normalization of pixel values to the range $[0, 1]$ helped stabilize the training process and improve convergence speed. In addition, the Adam optimizer enabled efficient parameter updates and contributed to the model's strong performance.

5 Conclusion

This project presented a convolutional neural network (CNN) model for recognizing handwritten Arabic digits. The system included data preprocessing, CNN architecture design, model training, and performance evaluation.

The proposed model was trained on a dataset containing 9,350 images of handwritten digits and evaluated on a test set of 1,870 images. The trained model achieved an overall accuracy of approximately **94.22%** on the test dataset.

These results demonstrate that convolutional neural networks provide an effective approach for handwritten digit recognition.

References

[1] H. Azis and I. N. Saly, "Handwritten Arabic Numerals (0–9) Image Dataset," Mendeley Data, 2024.

[2] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-Based Learning Applied to Document Recognition," Proceedings of the IEEE, 1998.

[3] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

[4] M. Abadi et al., "TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems," Google Research, 2015.